

STT 861: Theory of Probability & Statistics I

Lecture 4: Probability Calculus & Inequalities

Guanqun Cao

Department of Statistics & Probability
Michigan State University

Sep 10, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 1.2

Lecture 4 Outline

Today

- Consequences of the three axioms
- Inclusion–Exclusion principle
- Boole's inequality; Bonferroni's inequality

Elementary Probability Formulas

Propositions from the Axioms

- ① $P(\emptyset) = 0$
- ② $P(E) \leq 1$
- ③ $P(E^c) = 1 - P(E)$
- ④ $P(F - E) = P(FE^c) = P(F) - P(EF)$
- ⑤ $E \subset F \Rightarrow P(E) \leq P(F)$
- ⑥ $P(E \cup F) = P(E) + P(F) - P(EF)$

Proof of (3): $S = E \cup E^c$, $E \cap E^c = \emptyset$.

Axiom (A3): $P(S) = P(E) + P(E^c)$. Axiom (A2): $P(S) = 1$. Hence $P(E^c) = 1 - P(E)$. $\square$

Proof of (6): Write $E \cup F = E \cup (F \cap E^c)$ with these two sets disjoint.

$P(E \cup F) = P(E) + P(FE^c) = P(E) + P(F) - P(EF)$. $\square$

Inclusion–Exclusion Principle

General Inclusion–Exclusion

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_i P(E_i) - \sum_{i < j} P(E_i E_j) + \sum_{i < j < k} P(E_i E_j E_k) - \cdots + (-1)^{n+1} P(E_1 \cdots E_n)$$

For $n = 3$:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$$

Application: Subject Interests

In a class: $P(\text{Math}) = 0.6$, $P(\text{Stat}) = 0.4$, $P(\text{CS}) = 0.3$.

$P(\text{Math and Stat}) = 0.2$, $P(\text{Math and CS}) = 0.15$, $P(\text{Stat and CS}) = 0.1$,
 $P(\text{all three}) = 0.05$.

$P(\text{at least one}) = 0.6 + 0.4 + 0.3 - 0.2 - 0.15 - 0.1 + 0.05 = \mathbf{0.90}$

Boole's Inequality & Bonferroni's Inequality

Boole's Inequality (Union Bound)

For any events $E_1, \dots, E_n$:

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

Proof: Define $B_1 = E_1$, $B_k = E_k \cap E_1^c \cdots E_{k-1}^c$ (disjoint!).

Then $\bigcup E_i = \bigcup B_i$ (disjoint), so $P(\bigcup E_i) = \sum P(B_i) \leq \sum P(E_i)$ since $B_k \subset E_k$. $\square$

Bonferroni's Inequality

$$P\left(\bigcap_{i=1}^n E_i\right) \geq \sum_{i=1}^n P(E_i) - (n-1)$$

Proof: $P(\bigcap E_i) = 1 - P(\bigcup E_i^c) \geq 1 - \sum P(E_i^c) = 1 - \sum(1 - P(E_i)) = \sum P(E_i) - (n-1)$. $\square$

Practice with Probability Formulas

Example: Three Coin Tosses

Toss 3 fair coins. Let $A_i = \{\text{coin } i \text{ is H}\}$ for $i = 1, 2, 3$.

$$P(A_i) = 1/2 \text{ for each } i;$$

$$P(A_i \cap A_j) = 1/4 \text{ for } i \neq j;$$

$$P(A_1 \cap A_2 \cap A_3) = 1/8.$$

Find: $P(\text{at least one H}) = P(A_1 \cup A_2 \cup A_3)$

$$= 3(1/2) - 3(1/4) + 1(1/8) = 3/2 - 3/4 + 1/8 = \mathbf{7/8}$$

Check: $P(\text{no H}) = P(TTT) = 1/8; \quad 1 - 1/8 = 7/8. \checkmark$

Two Dice: Sum ≤ 4

S has 36 equally likely outcomes.

$$A = \{\text{sum} = 2\} = \{(1,1)\}, \quad B = \{\text{sum} = 3\} = \{(1,2), (2,1)\}, \quad C = \{\text{sum} = 4\} = \{(1,3), (2,2), (3,1)\}$$

$$P(\text{sum} \leq 4) = P(A) + P(B) + P(C) = 1/36 + 2/36 + 3/36 = \mathbf{6/36 = 1/6}$$

Continuity of Probability

Continuity from Axiom (A3)

(a) Increasing: If $E_1 \subset E_2 \subset E_3 \subset \dots$, then

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n)$$

(b) Decreasing: If $E_1 \supset E_2 \supset E_3 \supset \dots$, then

$$P\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n)$$

Example

$S = (0, 1]$, P is length. Let $E_n = (0, 1/n]$.

$E_1 \supset E_2 \supset \dots$; $P(E_n) = 1/n \rightarrow 0$.

$\bigcap_{n=1}^{\infty} E_n = \emptyset$, $P(\emptyset) = 0$. ✓

Classical Probability Examples

Example 1 (CB 1.2.5): Tossing 3 Coins

$S = \{HHH, THH, HTH, HHT, TTT, HTT, THT, TTH\}$, each with $P = 1/8$.

Let $A = \{\text{at least 2 heads}\} = \{HHH, THH, HTH, HHT\}$.

$$P(A) = 4/8 = 1/2.$$

Example 2: Two Dice Summing to 7

36 equally likely outcomes. $A = \{\text{sum}=7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$.

$$P(A) = 6/36 = 1/6.$$

Urn Sampling — Derivation from Axioms

Setup

w = number of white balls in urn

r = number of red balls in urn

$w + r$ = total number of balls

W_i = event of drawing the i -th white ball, $i = 1, 2, \dots, w$

R_j = event of drawing the j -th red ball, $j = 1, 2, \dots, r$

Assume $P(W_i) = P(R_j) = p$ for all i, j (symmetry).

Since $S = (\bigcup_{i=1}^w W_i) \cup (\bigcup_{j=1}^r R_j)$ and these events are disjoint:

$$P(S) = 1 = \sum_{i=1}^w P(W_i) + \sum_{j=1}^r P(R_j) = (w + r) p$$

$$\therefore p = \frac{1}{w + r}$$

Probability of White Ball

Let $W = W_1 \cup W_2 \cup \dots \cup W_w$.

$$P(W) = \sum_{i=1}^w P(W_i) = wp = \frac{w}{w + r}$$

Boole's Inequality (Union Bound)

Boole's Inequality

For any events $E_1, E_2, \dots, E_n$:

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

Proof. Define disjoint events:

$$B_1 = E_1, \quad B_2 = E_2 E_1^c, \quad B_3 = E_3 E_2^c E_1^c, \quad \dots, \quad B_n = E_n E_{n-1}^c \cdots E_1^c$$

① The $\{B_i\}$ are **pairwise disjoint** by construction.

② $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n B_i$, so by countable additivity:

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(B_i)$$

Bonferroni Inequality — Statement & Base Case

Bonferroni Inequality

For events $E_1, \dots, E_n$:

$$\sum_{i=1}^n P(E_i) - \sum_{i < j} P(E_i E_j) \leq P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$$

The **upper bound** is Boole's Inequality. The **lower bound** is proved by induction.

Proof of lower bound by induction.

Base case $n = 2$: Inclusion-exclusion gives

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 E_2) \geq P(E_1) + P(E_2) - P(E_1 E_2) \checkmark$$

Inductive hypothesis: Assume for $n - 1$ events:

$$\sum_{i=1}^{n-1} P(E_i) - \sum_{i < j \leq n-1} P(E_i E_j) \leq P\left(\bigcup_{i=1}^{n-1} E_i\right)$$

Bonferroni Inequality — Inductive Step (Part 1)

Inductive step. Define B_i as before and let $B = \bigcup_{i=1}^{n-1} B_i = \bigcup_{i=1}^{n-1} E_i$.

By inclusion-exclusion applied to B and E_n :

$$P\left(\bigcup_{i=1}^n E_i\right) = P(B \cup E_n) = P(B) + P(E_n) - P(B \cap E_n)$$

Applying the inductive hypothesis to $P(B)$:

$$P\left(\bigcup_{i=1}^n E_i\right) \geq \sum_{i=1}^{n-1} P(E_i) - \sum_{i < j \leq n-1} P(E_i \cap E_j) + P(E_n) - P(B \cap E_n) \quad (**)$$

Now we need to bound the last term $P(B \cap E_n)$. Expanding:

$$E_n B = E_n (B_1 \cup B_2 \cup \cdots \cup B_{n-1}) = E_n B_1 \cup E_n B_2 \cup \cdots \cup E_n B_{n-1}$$

Since the B_i are disjoint:

$$P(B \cap E_n) = \sum_{i=1}^{n-1} P(E_n B_i)$$

Bonferroni Inequality — Inductive Step (Part 2)

Bounding $P(E_n B_i)$. Recall $B_i = E_i E_{i-1}^c \cdots E_1^c \subseteq E_i$, so:

$$E_n B_i = E_n (E_i E_{i-1}^c \cdots E_1^c) \subseteq E_n E_i \quad \Rightarrow \quad P(E_n B_i) \leq P(E_n E_i)$$

Therefore:

$$P(B E_n) \leq \sum_{i=1}^{n-1} P(E_n E_i)$$

Substituting back into (**):

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &\geq \sum_{i=1}^n P(E_i) - \sum_{i < j \leq n-1} P(E_i E_j) - \sum_{i=1}^{n-1} P(E_n E_i) \\ &= \sum_{i=1}^n P(E_i) - \sum_{i < j \leq n} P(E_i E_j) \quad \square \end{aligned}$$

Remark

The last step collects the two sums: $\sum_{i < j \leq n-1} P(E_i E_j) + \sum_{i=1}^{n-1} P(E_n E_i) = \sum_{i < j \leq n} P(E_i E_j)$, completing the induction. □

Activity 4 – Probability Calculus

Problem 1. A student passes Course A with probability 0.8, Course B with probability 0.7. The probability of passing both is 0.6. Find:

- (a) $P(A \cup B)$
- (b) $P(A^c \cap B^c)$ (fails both)
- (c) $P(\text{passes exactly one})$

Problem 2. Three events E_1, E_2, E_3 have $P(E_i) = 0.7$ for each i .

- (d) Use Boole's inequality to bound $P(E_1 \cup E_2 \cup E_3)$.
- (e) Use Bonferroni's inequality to give a lower bound for $P(E_1 \cap E_2 \cap E_3)$.

Problem 3. Toss a fair coin 4 times. Let $A_i = \{i\text{-th toss is H}\}$.

- (f) Find $P(\text{at least one tail})$ using inclusion-exclusion (or complement).

Activity 4 – Solutions

Problem 1:

- (a) $P(A \cup B) = 0.8 + 0.7 - 0.6 = \mathbf{0.9}$
- (b) $P(A^c \cap B^c) = P((A \cup B)^c) = 1 - 0.9 = \mathbf{0.1}$
- (c) $P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.9 - 0.6 = \mathbf{0.3}$

Problem 2:

- (d) Boole: $P(E_1 \cup E_2 \cup E_3) \leq 3 \times 0.7 = \mathbf{2.1}$ (trivially ≤ 1 , so bound is 1)
- (e) Bonferroni: $P(E_1 \cap E_2 \cap E_3) \geq 3(0.7) - 2 = 2.1 - 2 = \mathbf{0.1}$

Problem 3:

- (f) $P(\text{at least one T}) = 1 - P(HHHH) = 1 - 1/16 = \mathbf{15/16}$